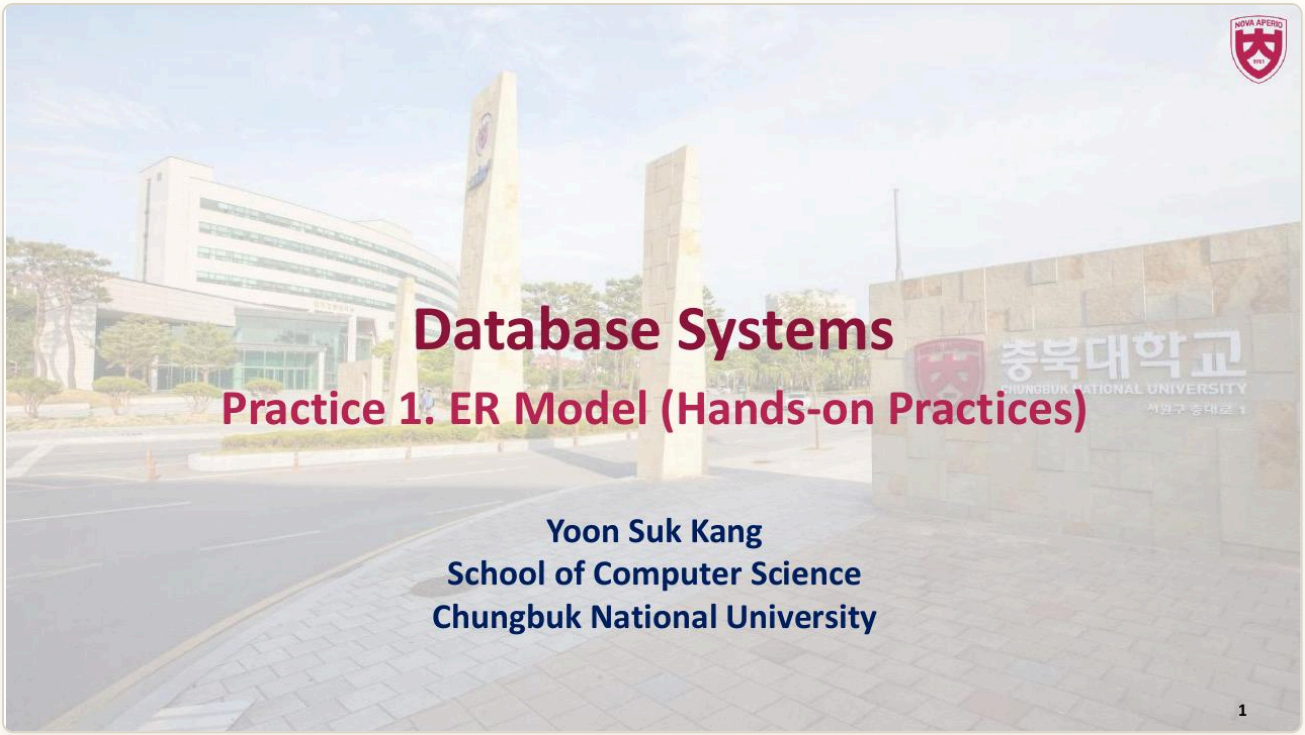
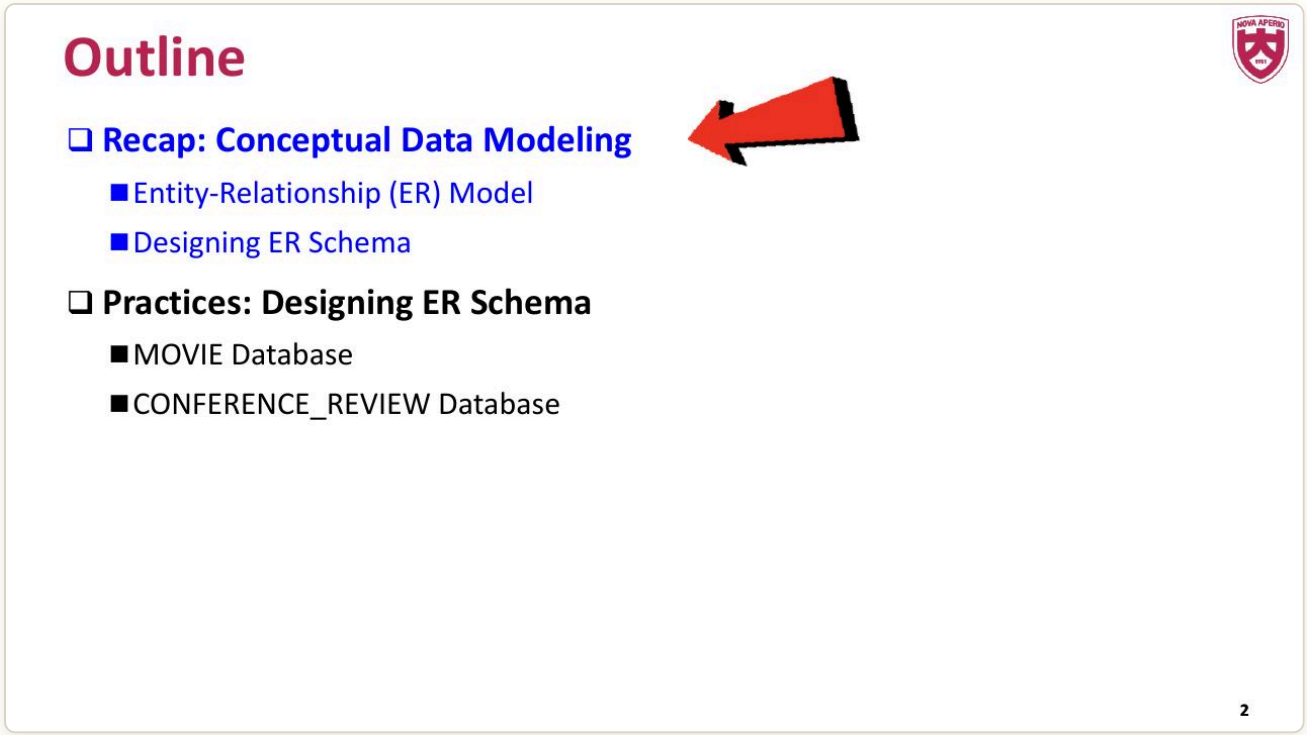


데이터베이스 시스템 — 실습 1: ER 모델

- Yoon Suk Kang · 충북대 소프트웨어학부

✍️ 메모

💡 이해 요구사항 문장 → ER 다이어그램 연습. 시험 서술형(ER 그리기) 대비 1순위

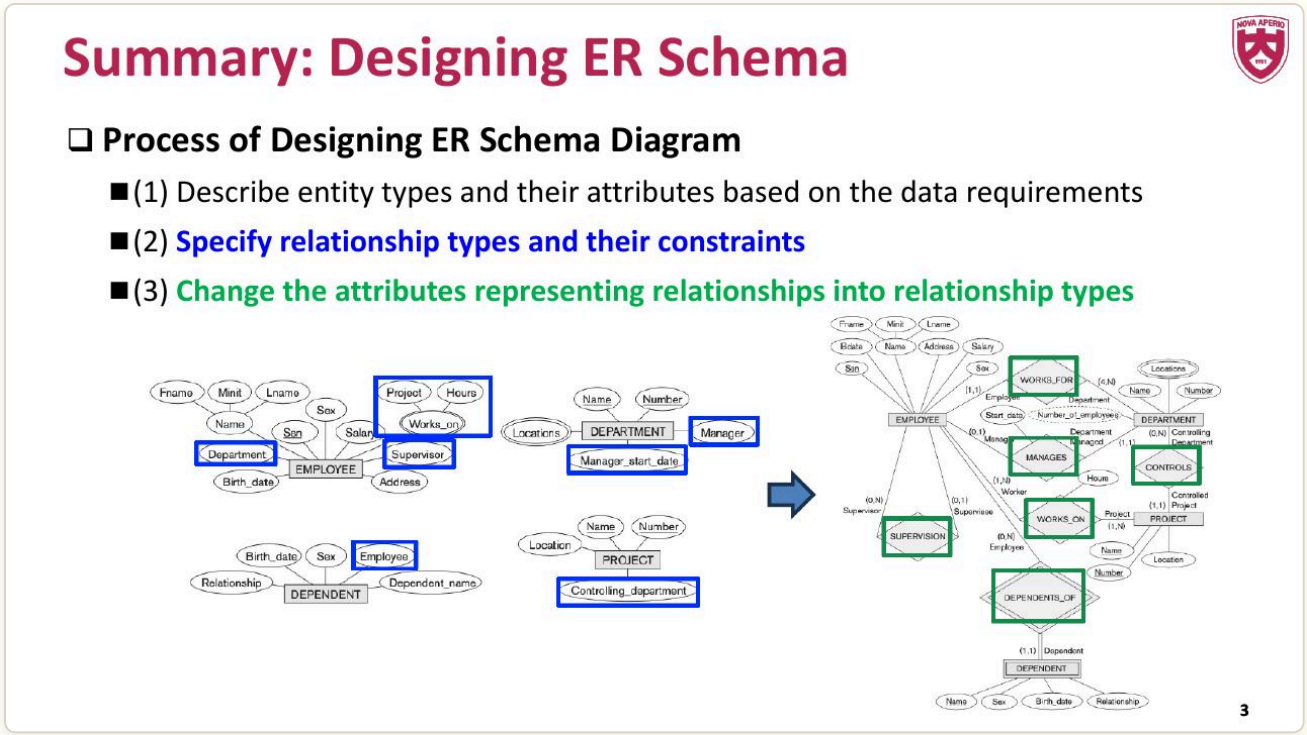


목차 (Outline)

- **복습: 개념적 데이터 모델링** — ER 모델 / ER 스키마 설계
- **실습: ER 스키마 설계** — MOVIE DB / CONFERENCE_REVIEW DB

✍ 메모

— 넘어가도 되는 쪽

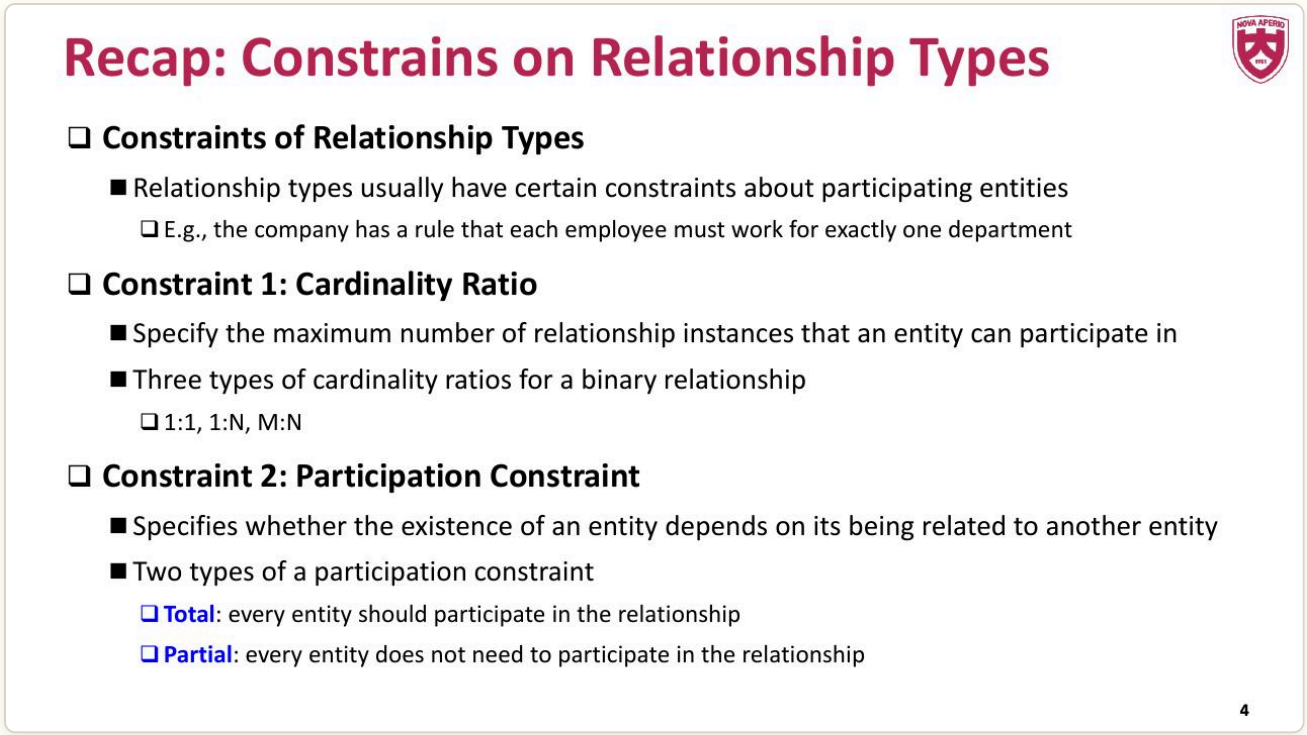


요약: ER 스키마 설계

- ER 스키마 다이어그램 설계 과정
  - (1) 요구사항에 따라 **엔티티 타입과 속성** 기술
  - (2) **관계 타입과 제약** 지정
  - (3) 관계를 나타내던 **속성을 관계 타입으로** 바꾸기

✍ 메모

★ 시험 아래 두 실습 모두 이 Step 1 → 2 → 3 순서

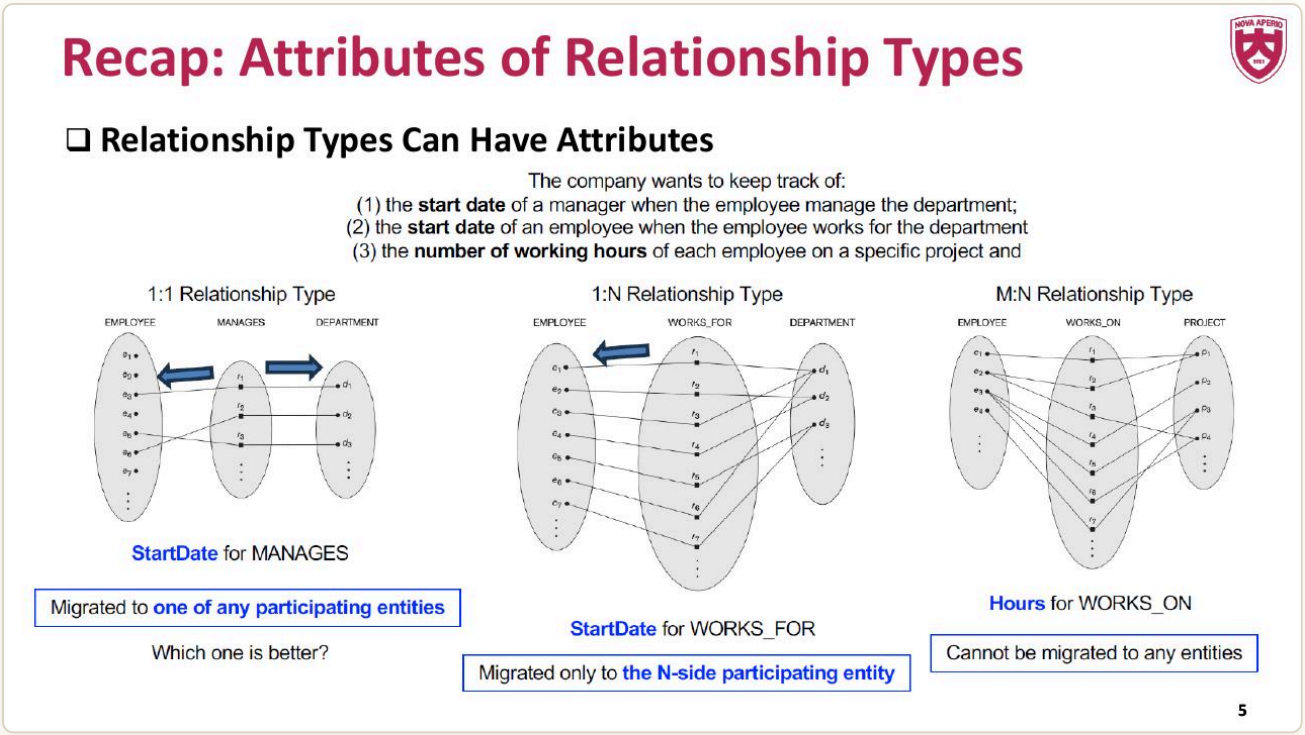


복습: 관계 타입의 제약

- 관계 타입에는 보통 참여 엔티티에 대한 제약이 있다
  - 예: 회사 규칙 — 직원은 정확히 한 부서에서 일한다
- **제약 1: Cardinality Ratio** — 엔티티가 참여할 수 있는 관계 인스턴스의 **최대** 수
  - 이항 관계: 1:1, 1:N, M:N
- **제약 2: Participation Constraint** — 엔티티의 존재가 다른 엔티티와의 관계에 **의존**하는지
  - **Total**: 모든 엔티티가 참여 / **Partial**: 모두 참여할 필요 없음

✍ 메모

— 넘어가도 되는 쪽

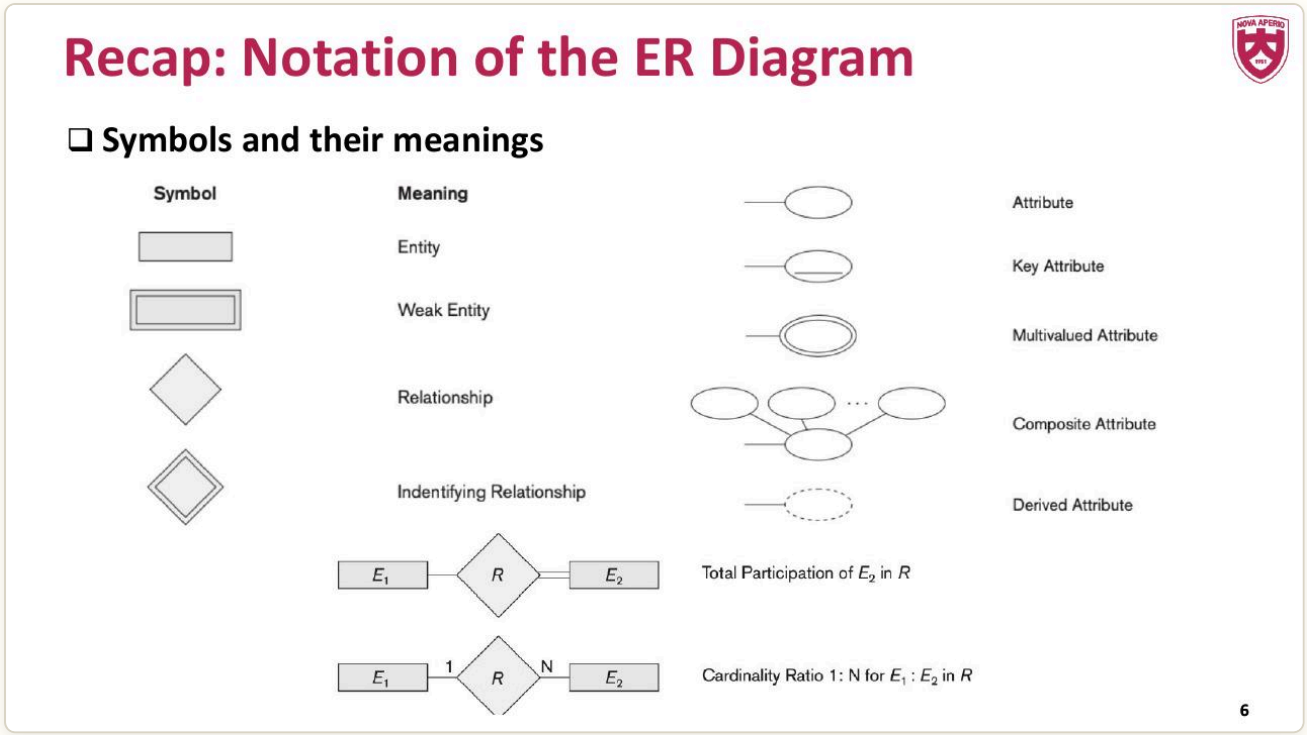


복습: 관계 타입의 속성

- 관계 타입도 속성을 가질 수 있다
  - MANAGES의 StartDate, WORKS_ON의 Hours

✍ 메모

★ 시험 M:N 관계의 속성은 관계에 남는다 — 이번 실습의 Role·Quotes·Rating이 그 예

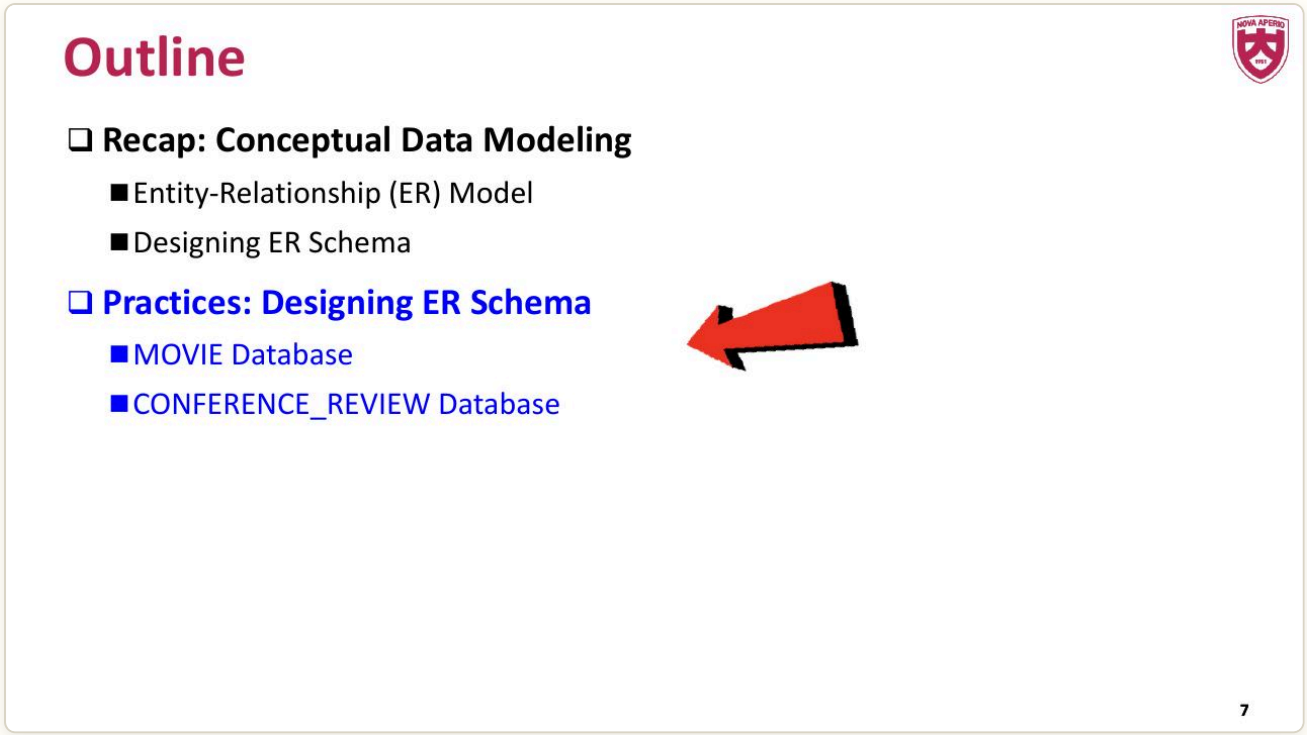


복습: ER 다이어그램 기호

- 사각형 = 엔티티 / 이중 사각형 = 약한 엔티티
- 마름모 = 관계 / 이중 마름모 = 식별 관계
- 타원 = 속성 / 밑줄 = 키 / 이중 타원 = 다중값 / 타원에 타원 = 복합 / 점선 타원 = 유도
- 이중선 = 전체 참여 / 1, N = 카디널리티 / (min, max) = 구조적 제약

✍ 메모

★ 시험 기호표는 2강 p.50·p.70과 같다



목차 (Outline)

- **복습: 개념적 데이터 모델링** — ER 모델 / ER 스키마 설계
- **실습: ER 스키마 설계** — MOVIE DB / CONFERENCE_REVIEW DB

✍️ 메모

💡 이해 실습 1: MOVIE

Exercise 1: ER Schema Diagram

❑Designing ER Diagram for MOVIE Database

■Consider a MOVIE database in which data is recorded about the movie industry

■The data requirements are summarized as follows:



❑MOVIE

- ■ Each movie is identified by (**title and year of release**)
- ■ Each has a length in minutes and a plot outline
- ■ Each is classified under one or more genres (e.g., horror, action, drama, ...)
- ■ Each movie has a production company, one or more directors, and one or more actors appearing in it



❑ACTOR

- ■ Actors are identified by (**name and date of birth**) and appear in one or more movies
- ■ Each actor has a role in the movie and has zero or more quotable quotes to speak





8

연습 1: MOVIE DB ① 요구사항

- 영화 산업 데이터를 기록하는 MOVIE DB
- **MOVIE**
  - (제목, 개봉 연도)로 식별
  - 상영 시간(분)과 줄거리 요약이 있음
  - 장르 **하나 이상**으로 분류 (공포, 액션, 드라마 ...)
  - 제작사 하나, 감독 **한 명 이상**, 배우 **한 명 이상**
- **ACTOR**
  - (이름, 생년월일)로 식별, 영화 **하나 이상**에 출연
  - 영화마다 배역(role)이 있고, 명대사(quotes)가 **0개 이상**

✍ 메모

💡 이해 '(title and year)로 식별' → 복합 키 / 'one or more genres' → 다중값

Exercise 1: ER Schema Diagram (cont.)

❑Designing ER Diagram for MOVIE Database

■Consider a MOVIE database in which data is recorded about the movie industry

■The data requirements are summarized as follows:



❑**DIRECTOR**

■ Directors are identified by (**name and date of birth**) and direct one or more movies

■ It is possible for a director to act in a movie (including one that she/he may also direct)



❑**PRODUCTION**

■ Production companies are identified by (**name**) and each has an address

■ A production company produces one or more movies





9

연습 1: MOVIE DB ② 요구사항 (계속)

- **DIRECTOR**
  - (이름, 생년월일)로 식별, 영화 **하나 이상**을 감독
  - 감독이 영화에 출연할 수도 있음 (자기가 감독한 영화 포함)
- **PRODUCTION**
  - 제작사는 (이름)으로 식별, 주소가 있음
  - 제작사는 영화를 **하나 이상** 제작

✍️ 메모

⚠ 주의 감독도 출연 가능 → 감독–영화 사이 '출연' 관계가 따로 필요 (DIR_ACTS)

Step 1: Describing the Entity Types

❑ Entity Type 1: MOVIE

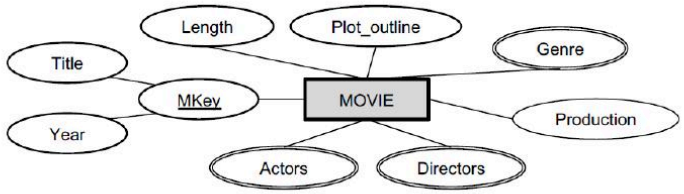


■ Requirements:

- ❑ Each movie is identified by (**title and year of release**)
- ❑ Each has a length in minutes and a plot outline
- ❑ Each is classified under one or more genres (e.g., horror, action, drama, ...)
- ❑ Each movie has a production company, one or more directors, and one or more actors appearing in it

■ Attributes:

- ❑ *Mkey(Title, Year), Length, Plot_outline, {Genre}, Production, {Directors}, {Actors}*



10

Step 1: 엔티티 타입 — MOVIE

- 요구사항: (제목, 연도)로 식별 / 상영 시간, 줄거리 / 장르 하나 이상 / 제작사, 감독들, 배우들
- 속성:

```
Mkey(Title, Year), Length, Plot_outline, {Genre}, Production, {Directors}, {Actors}
```

✍ 메모

★ 시험 { } = 다중값, ( ) = 복합. Mkey는 복합 키

💡 이해 Production·Directors·Actors는 다른 엔티티를 가리키는 속성 → Step 2에서 관계로 바뀜

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 2: ACTOR**

■ Requirements:

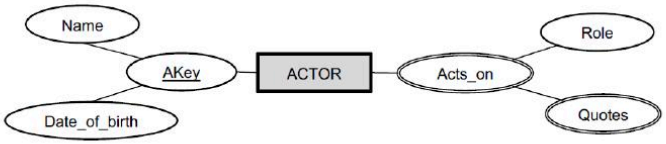
❑ Actors are identified by (**name and date of birth**) and appear in one or more movies

❑ Each actor has a role in the movie and has zero or more quotable quotes to speak



■ Attributes:

❑ *Akey(Name, Date_of_birth), {Acts_on(Movie, Role, {Quotes})}*



11

Step 1: 엔티티 타입 — ACTOR

- 요구사항: (이름, 생일)로 식별, 영화 하나 이상 출연 / 배역과 명대사 0개 이상
- 속성:

```
Akey(Name, Date_of_birth), {Acts_on(Movie, Role, {Quotes})}
```

✍ 메모

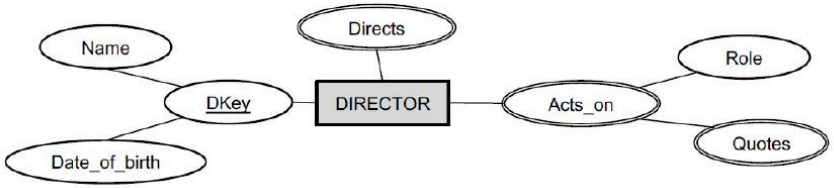
✏ 예·풀이 Acts_on = 다중값 + 복합, 그 안에 Quotes도 다중값

Step 1: Describing the Entity Types (cont.)

❑ Entity Type 3: DIRECTOR



- ■ Requirements:
  - ❑ Directors are identified by (**name and date of birth**) and direct one or more movies
  - ❑ It is possible for a director to act in a movie (including one that she/he may also direct)
- ■ Attributes
  - ❑ *Dkey(Name, Date_of_birth), {Directs}, {Acts_on(Movie, Role, {Quotes})}*



12

Step 1: 엔티티 타입 — DIRECTOR

- 요구사항: (이름, 생일)로 식별, 영화 하나 이상 감독 / 영화에 출연할 수도
- 속성:

```
Dkey(Name, Date_of_birth), {Directs}, {Acts_on(Movie, Role, {Quotes})}
```

✍ 메모

— 넘어가도 되는 쪽

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 4: PRODUCTION**

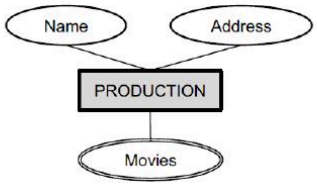
■ Requirements:

❑ Production companies are identified by (**name**) and each has an address

❑ A production company produces one or more movies

■ Attributes:

❑ *Name, Address, {Movies}*



13

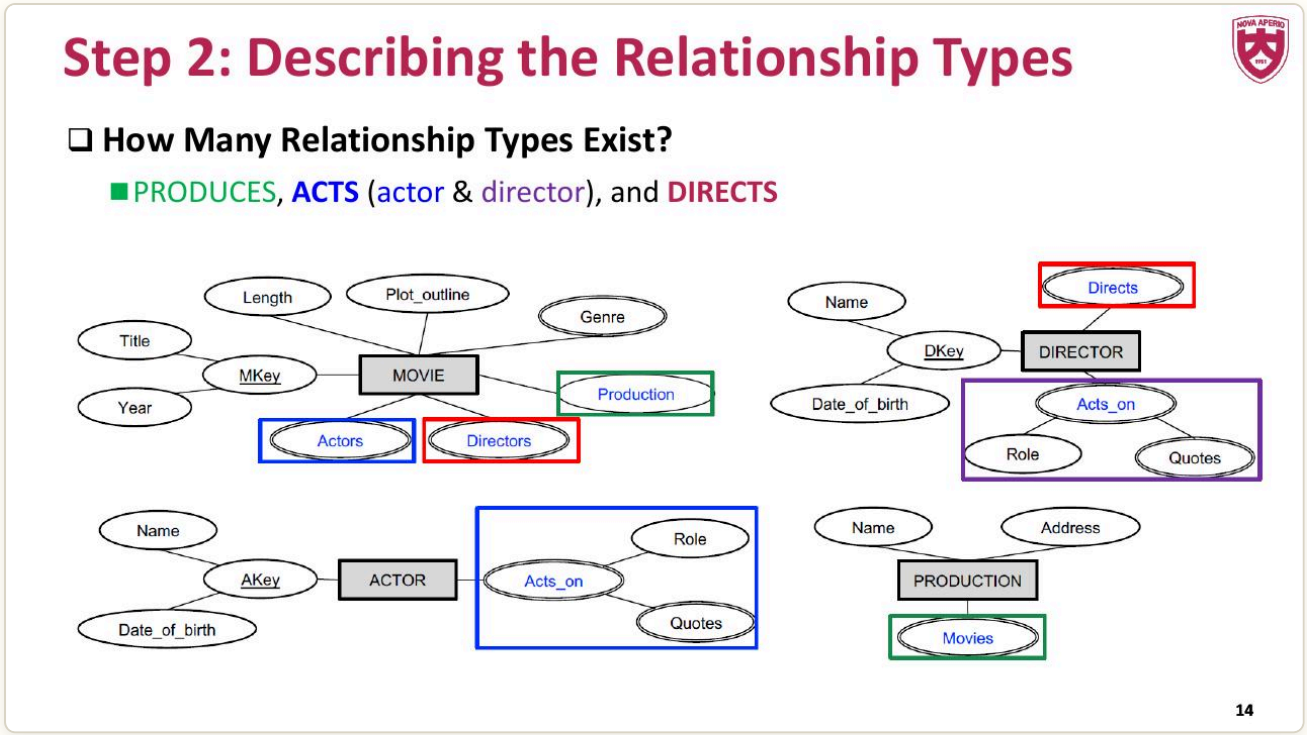
Step 1: 엔티티 타입 — PRODUCTION

- 요구사항: (이름)으로 식별, 주소 / 영화 하나 이상 제작
- 속성:

```
Name, Address, {Movies}
```

✍ 메모

— 넘어가도 되는 쪽



Step 2: 관계 타입은 몇 개?

- **PRODUCES**, **ACTS**(배우 & 감독), **DIRECTS**

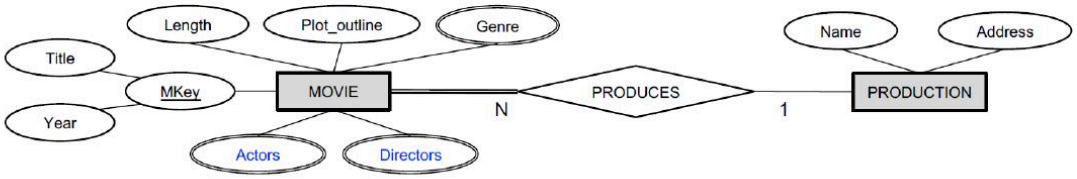
✍ 메모

💡 이해 ACTS는 배우용(ACTS)과 감독용(DIR_ACTS) 두 개로 그려짐 (p.18)

Step 2: Describing the Relationship Types

❑ Relationship Type 1: PRODUCES

- ■ Relating **PRODUCTION** and **MOVIE** entity types
  - ❑ A single production company can produce many movies, but a movie has to be produced by a single production company
    - ■ i.e., Cardinality ratio of this relationship type: (1:N)
  - ❑ The MOVIE entity type has a total participation constraint



15

관계 1: PRODUCES

- PRODUCTION과 MOVIE를 잇는다
  - 제작사 하나가 영화 여러 편을 제작, 영화는 제작사 **하나**가 제작
  - 카디널리티: **1:N**
  - MOVIE는 **전체 참여**

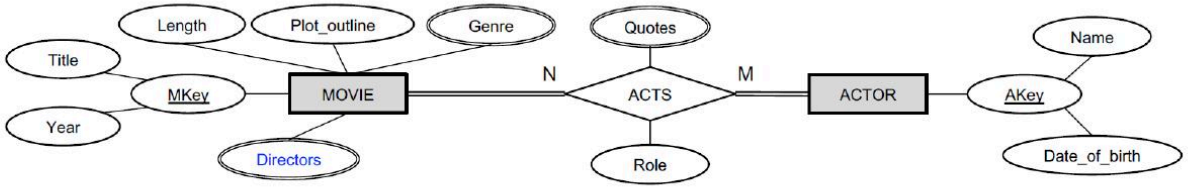
✍ 메모

★ 시험 PRODUCES 1:N, MOVIE total

Step 2: Describing the Relationship Types (cont.)

❑ Relationship Type 2: ACTS

- ■ Relating **ACTOR** and **MOVIE** entity types
  - ❑ An actor can act in many movies, and a movie can have many actors
    - ■ i.e., Cardinality ratio of this relationship type: (M:N)
  - ❑ This relationship type has two relationship attributes *Role* and *Quotes* (the multivalued attribute)
    - ■ The *Role* attribute names a role that the actor plays in the movie, and the *Quotes* are quotes spoken by an actor
  - ❑ The both entity types have a total participation constraint



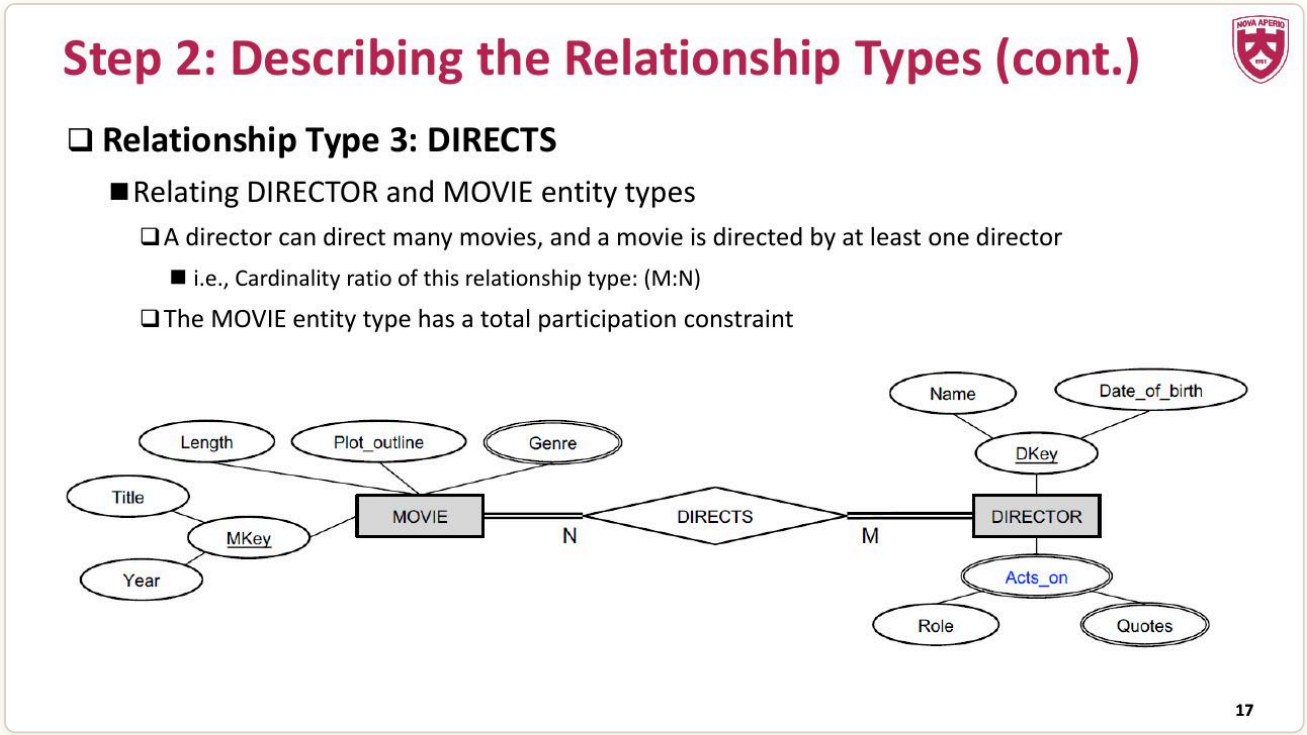
16

관계 2: ACTS

- ACTOR와 MOVIE를 잇는다
  - 배우는 여러 영화에, 영화에는 여러 배우 — **M:N**
  - 관계 속성 2개: **Role**, **Quotes**(다중값)
    - Role = 배우가 그 영화에서 맡은 역, Quotes = 배우가 한 대사
  - **양쪽 모두 전체 참여**

✍ 메모

★ 시험 Role·Quotes는 배우–영화 조합으로 정해짐 → 관계 속성 (M:N이라 엔티티로 못 옮김)

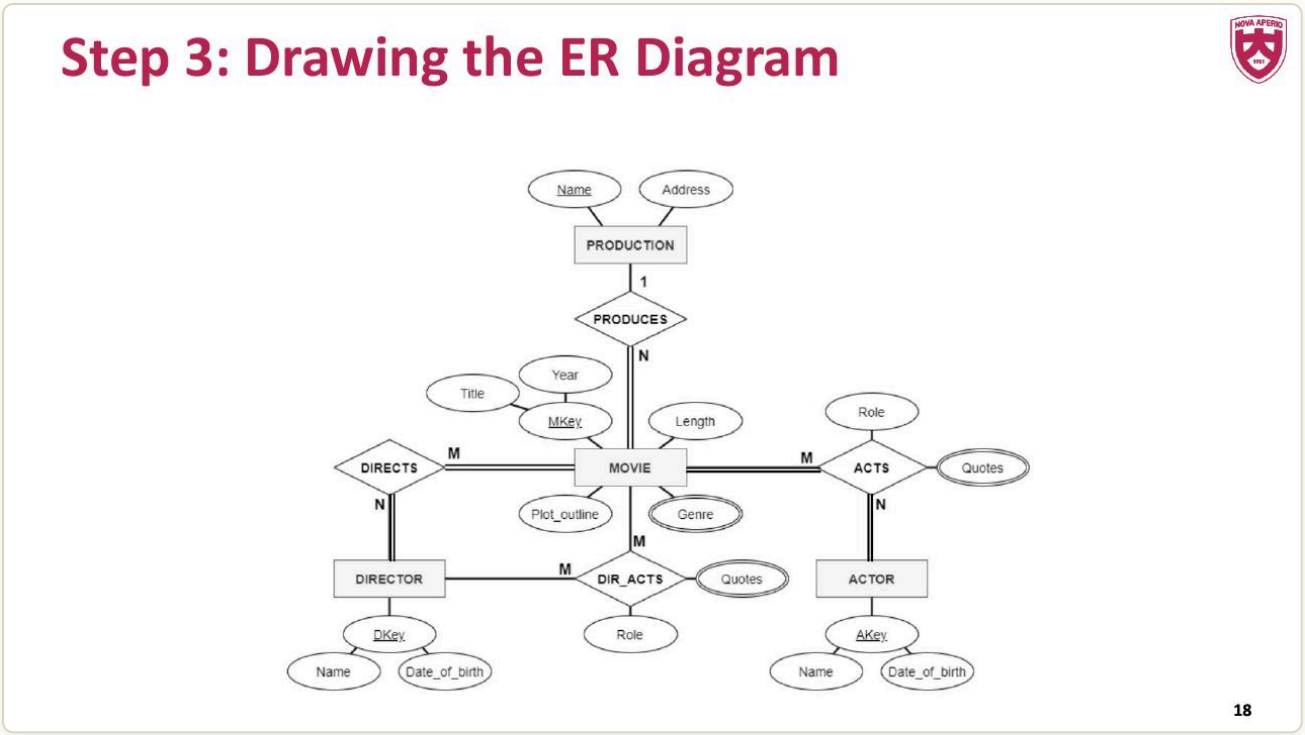


관계 3: DIRECTS

- DIRECTOR와 MOVIE를 잇는다
  - 감독은 여러 영화를 감독, 영화는 감독 **최소 한 명** — **M:N**
  - MOVIE는 **전체 참여**

✍ 메모

⚠ 주의 그림(p.18)은 DIRECTOR 쪽도 이중선(감독은 영화 하나 이상)



Step 3: ER 다이어그램 (MOVIE)

- PRODUCTION(Name, Address) —1— PRODUCES —N═ MOVIE
- MOVIE(MKey(Title, Year), Length, Plot_outline, {Genre})
- MOVIE ═M— ACTS —N═ ACTOR, 관계 속성 Role, {Quotes}
- MOVIE ═M— DIRECTS —N═ DIRECTOR
- MOVIE —M— DIR_ACTS —M— DIRECTOR, 관계 속성 Role, {Quotes}
- ACTOR(AKey(Name, Date_of_birth)), DIRECTOR(DKey(Name, Date_of_birth))

✍ 메모

★ 시험 정답 외우기: PRODUCES 1:N(MOVIE total) · ACTS M:N(Role, {Quotes}, 양쪽 total) · DIRECTS M:N(MOVIE total)

✏ 예·풀이 DIR_ACTS = 감독이 출연하는 경우 — 양쪽 partial

Exercise 2: Conference Review Database

❑ Background

- ■ Consider a CONFERENCE REVIEW database in which researchers submit their research papers for publication
- ■ Reviews by reviewers are recorded for use in the paper selection process
- ■ The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper



19

연습 2: CONFERENCE REVIEW DB — 배경

- 연구자들이 논문을 제출하는 학회 심사 DB
  - 심사위원의 심사가 기록되어 논문 선정에 쓰임
  - 주 사용자는 심사위원 — 논문마다 평가 질문에 답하고, 채택/거절 추천을 남김

✍ 메모

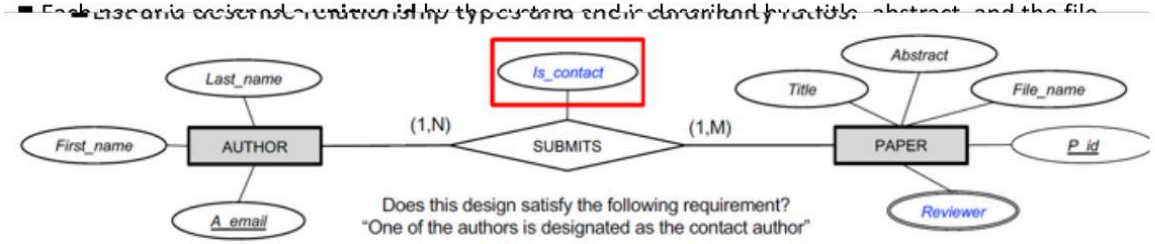
— 넘어가도 되는 쪽

Data Requirements

❑ **AUTHOR**

- Authors of papers are uniquely identified by **e-mail id** and have the first and last names
- Authors can submit one or more papers

❑ **PAPER**



- A reviewer rates each paper assigned to her/him on a scale of **1 to 10** in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference
- Each reviewer provides an overall recommendation to each paper assigned to her/him
- Each review **contains two types of written comments**: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)

20

연습 2: 데이터 요구사항

- **AUTHOR**: 이메일로 유일하게 식별, 이름·성 / 논문을 하나 이상 제출
- **PAPER**: 시스템이 유일한 id 부여, 제목·초록·파일 이름
  - 저자 여러 명 가능, 그중 한 명이 **교신 저자(contact author)**
  - 논문 하나에 심사위원 **2~4명** 배정
- **REVIEWER**: 이메일로 유일하게 식별, 이름·성·전화·소속·관심 주제
  - 배정된 논문을 4개 항목에 1~10점으로 평가: (1) 기술적 가치 (2) 가독성 (3) 독창성 (4) 학회 관련성
  - 논문마다 종합 추천
  - 심사평 두 종류: (1) 심사위원회만 보는 것 (2) 저자에게 주는 피드백

✍ 메모

💡 이해 숫자 표현('2~4명', '하나 이상')은 (min, max)로 바로 옮긴다

Step 1: Describing the Entity Types

❑ **Entity Type 1: AUTHOR**

■ Requirements:

❑ Authors of papers are uniquely identified by **e-mail id** and have the first and last names

❑ Authors can submit one or more papers

■ (Paper) Each can have multiple authors, but one of the authors is designated as the contact author

■ Attributes (List entity types and their attributes based on the requirements):



21

Step 1: AUTHOR (문제)

- 요구사항을 보고 AUTHOR의 속성을 적어라

✍ 메모

✏ 예·풀이 답은 p.29

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 2: PAPER**

■ Requirements:

- ❑ Each paper is assigned a **unique id** by the system and described by a title, abstract, and the file name
- ❑ Each can have multiple authors, but one of the authors is designated as the contact author
- ❑ A paper is assigned between two and four reviewers

■ Attributes (List entity types and their attributes based on the requirements):



22

Step 1: PAPER (문제)

- 요구사항을 보고 PAPER의 속성을 적어라

✍ 메모

✏ 예·풀이 답은 p.30

Step 1: Describing the Entity Types (cont.)

❑ Entity Type 3: REVIEWER

■ Requirements:

- ❑ Reviewers of papers are uniquely identified by **e-mail id**
- ❑ Each reviewer has the first and last name, phone number, affiliation, and topics of interest
- ❑ A reviewer rates each paper assigned to her/him on a scale of 1 to 10 in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference
- ❑ Each reviewer provides an overall recommendation to each paper assigned to her/him
- ❑ Each review contains two types of written comments: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)

■ Attributes (List entity types and their attributes based on the requirements):

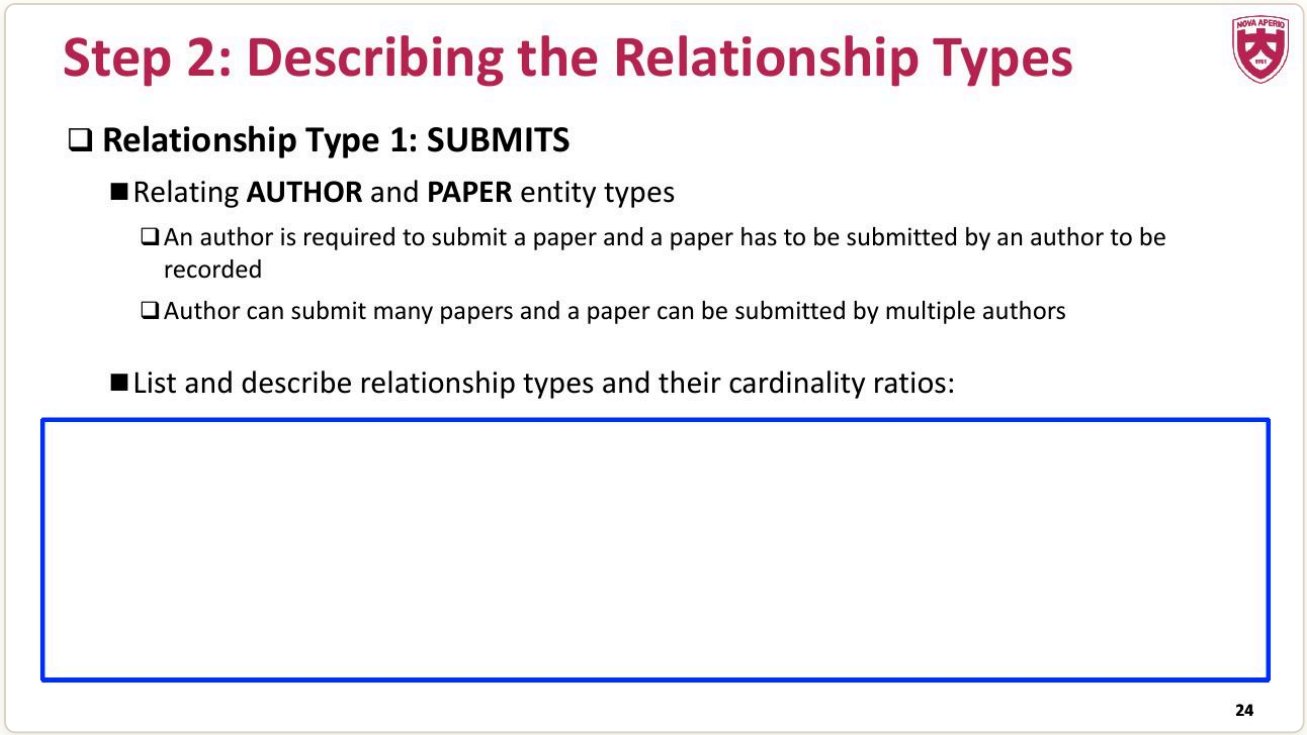
23

Step 1: REVIEWER (문제)

- 요구사항을 보고 REVIEWER의 속성을 적어라

✍ 메모

✏ 예·풀이 답은 p.31~32

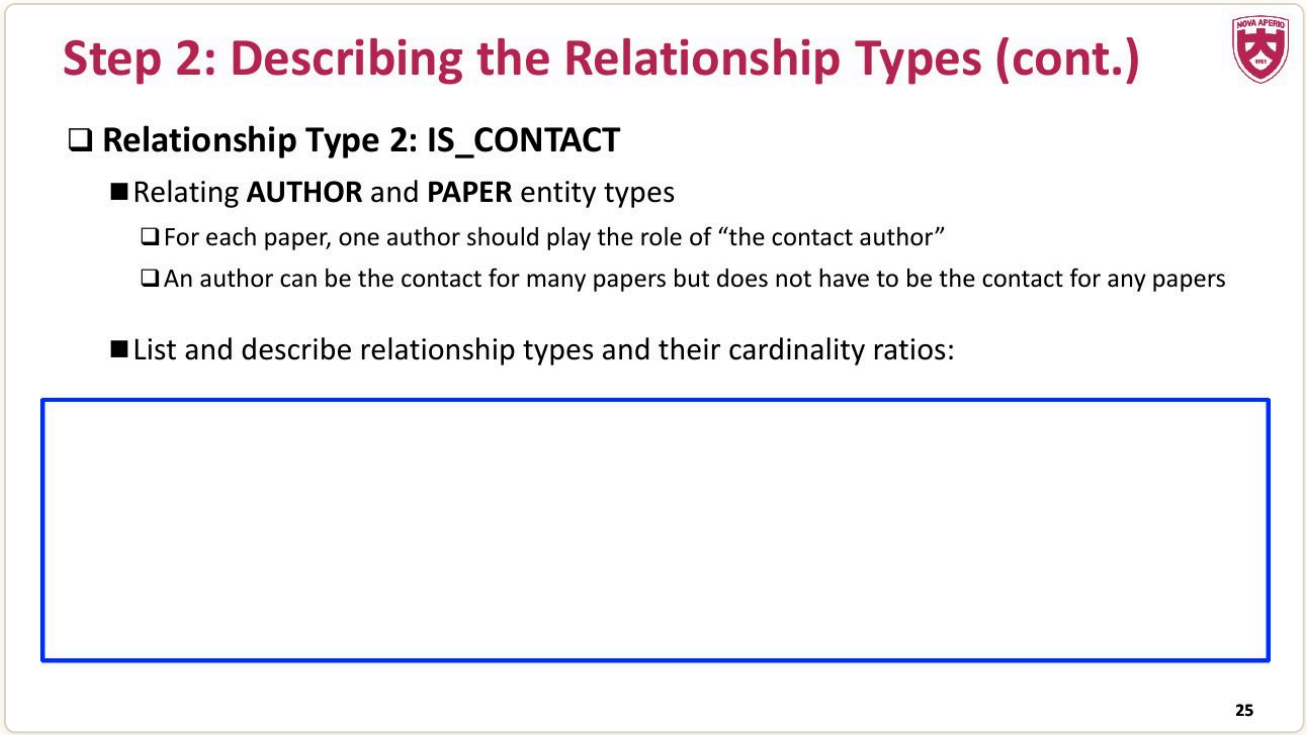


Step 2: SUBMITS (문제)

- AUTHOR와 PAPER를 잇는 관계
  - 저자는 논문을 제출해야 기록되고, 논문은 저자가 제출해야 기록된다
  - 저자는 여러 논문을, 논문은 여러 저자가 제출 가능

✍ 메모

✏ 예·풀이 답은 p.33

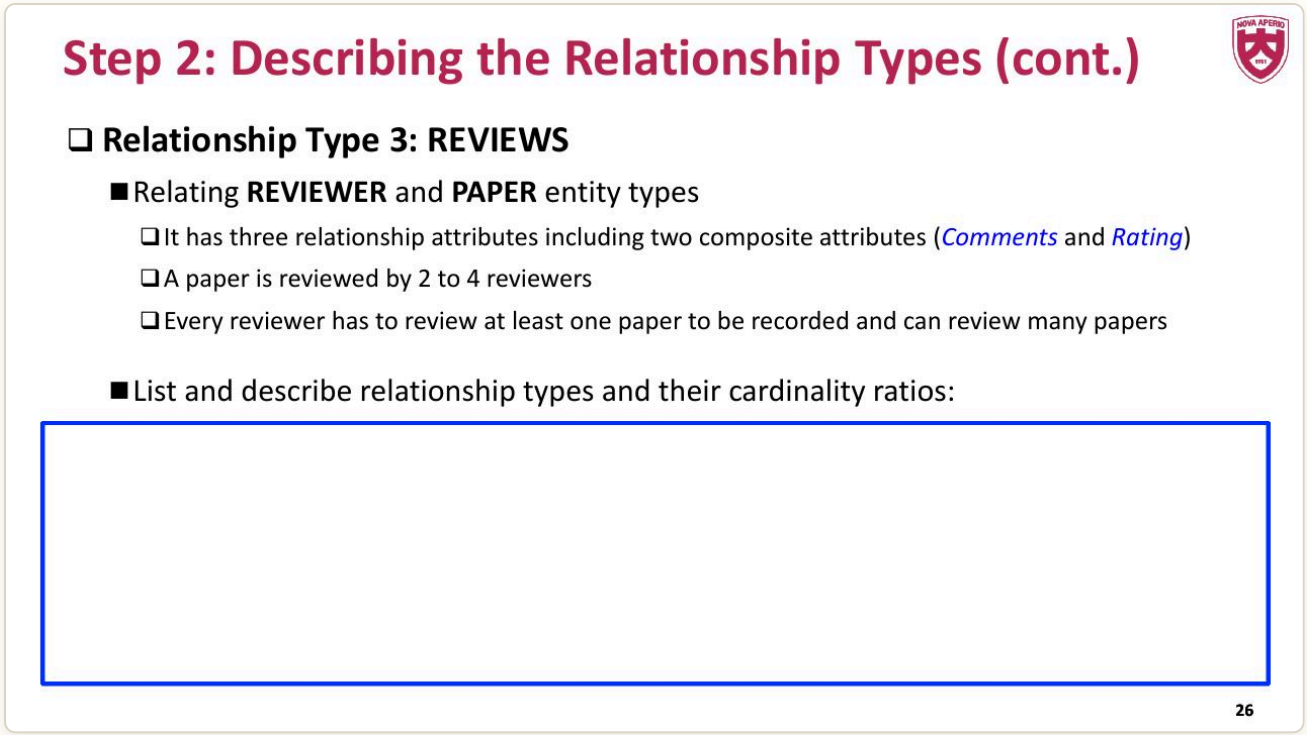


Step 2: IS_CONTACT (문제)

- AUTHOR와 PAPER를 잇는 관계
  - 논문마다 저자 한 명이 '교신 저자' 역할
  - 저자는 여러 논문의 교신 저자일 수 있지만, 아무 논문의 교신 저자가 아니어도 됨

✍ 메모

✏ 예·풀이 답은 p.34

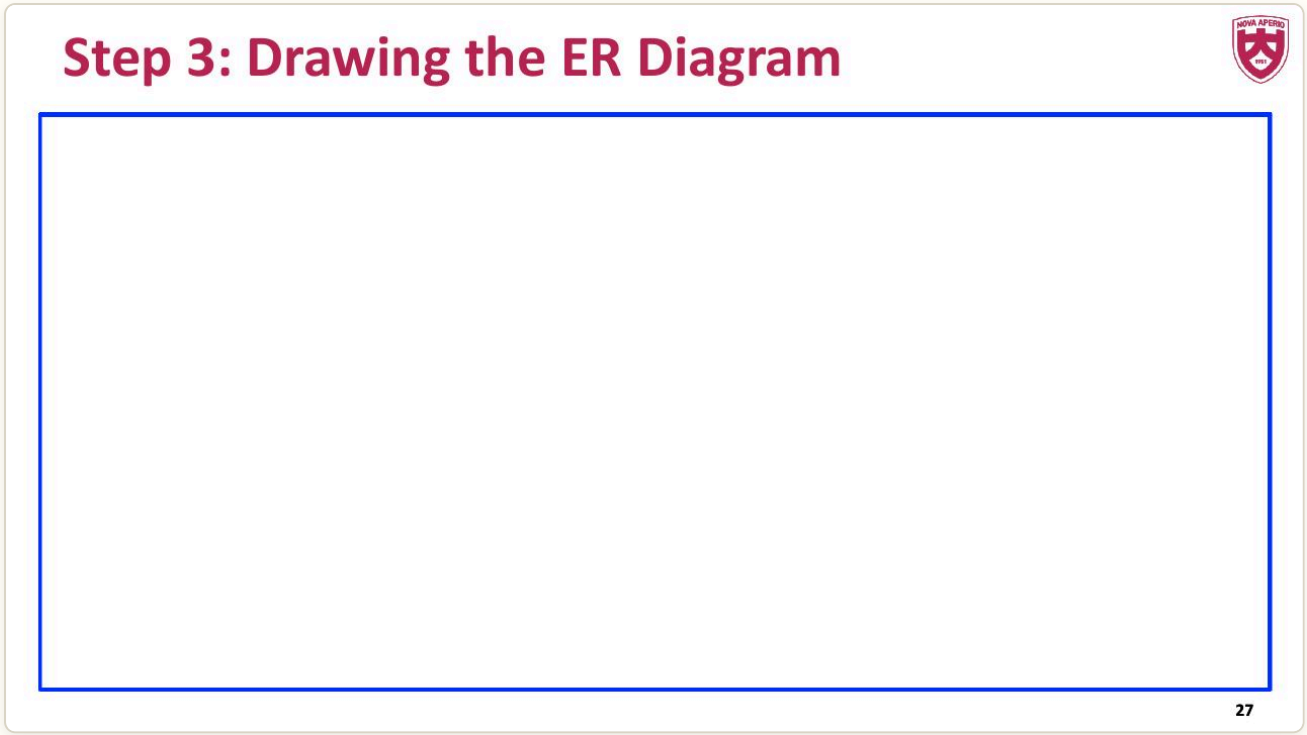


Step 2: REVIEWS (문제)

- REVIEWER와 PAPER를 잇는 관계
  - 관계 속성 3개 — 그중 복합 속성 2개(Comments, Rating)
  - 논문은 심사위원 2~4명이 심사
  - 심사위원은 최소 한 편을 심사해야 기록되고, 여러 편 심사 가능

✍ 메모

✏ 예·풀이 답은 p.35



Step 3: ER 다이어그램 (문제)

- (빈 그림 — 직접 그려 보기)

✍️ 메모

💡 이해 답 보기 전에 손으로 그려 보고 p.36과 비교



Q&A

- ER Model (Hands-on Practices) — 끝
- p.29~36은 연습 2 정답

✍ 메모

— 넘어가도 되는 쪽

Step 1: Describing the Entity Types

❑ Entity Type 1: AUTHOR

■ Requirements:

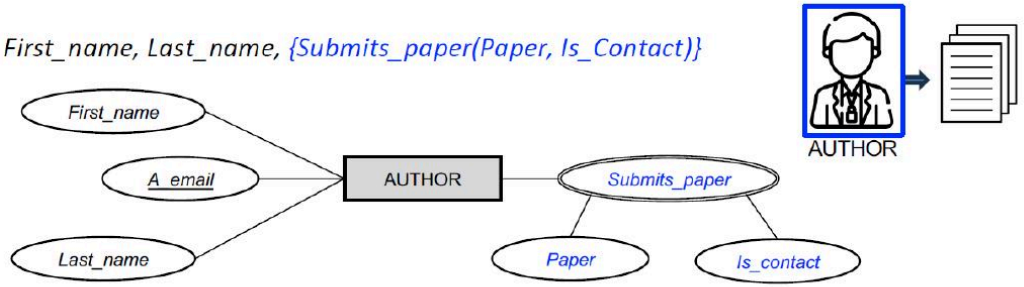
❑ Authors of papers are uniquely identified by **e-mail id** and have the first and last names

❑ Authors can submit one or more papers

■ (Paper) Each can have multiple authors, but one of the authors is designated as the contact author

■ Attributes (List entity types and their attributes based on the requirements):

A_email, First_name, Last_name, {Submits_paper(Paper, Is_Contact)}



29

정답 Step 1: AUTHOR

```
A_email, First_name, Last_name, {Submits_paper(Paper, Is_Contact)}
```

- A_email이 키 (밑줄)
- Submits_paper = 다중값 + 복합 (논문, 교신 저자 여부)

✍ 메모

💡 이해 Submits_paper는 Step 2에서 SUBMITS·IS_CONTACT 관계로 바뀐다

Step 1: Describing the Entity Types (cont.)

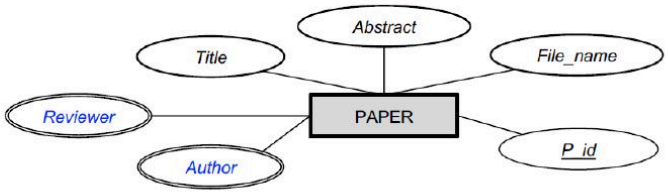
❑ Entity Type 2: PAPER

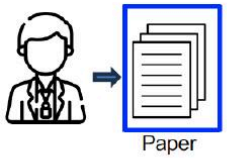
■ Requirements:

- ❑ Each paper is assigned a **unique id** by the system and described by a title, abstract, and the file name
- ❑ Each can have multiple authors, but one of the authors is designated as the contact author
- ❑ A paper is assigned between two and four reviewers

■ Attributes (List entity types and their attributes based on the requirements):

P_id, Title, Abstract, File_name , {Author}, {Reviewer}





30

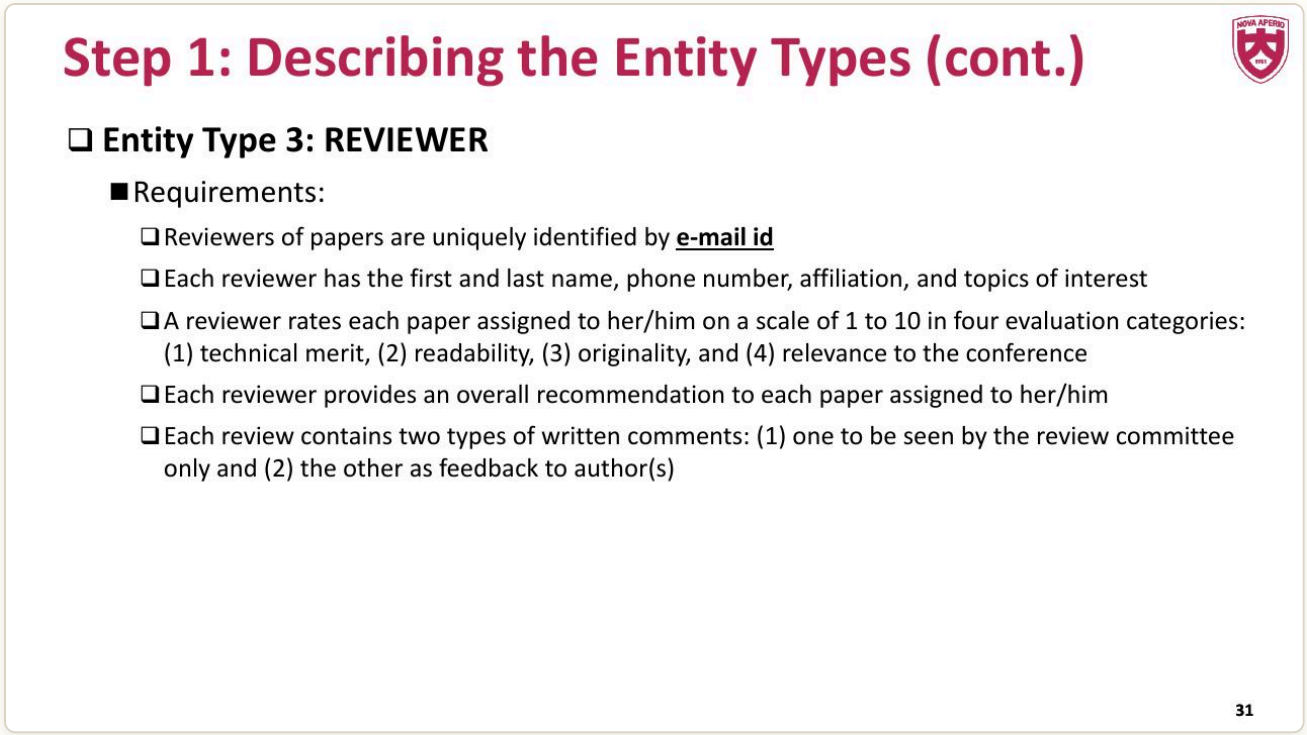
정답 Step 1: PAPER

```
P_id, Title, Abstract, File_name, {Author}, {Reviewer}
```

- P_id가 키, Author·Reviewer는 다중값

✍ 메모

— 넘어가도 되는 쪽



정답 Step 1: REVIEWER — 요구사항

- 이메일로 식별 / 이름·성·전화·소속·관심 주제
- 배정 논문을 4개 항목 1~10점 평가 / 종합 추천 / 심사평 두 종류

✍ 메모

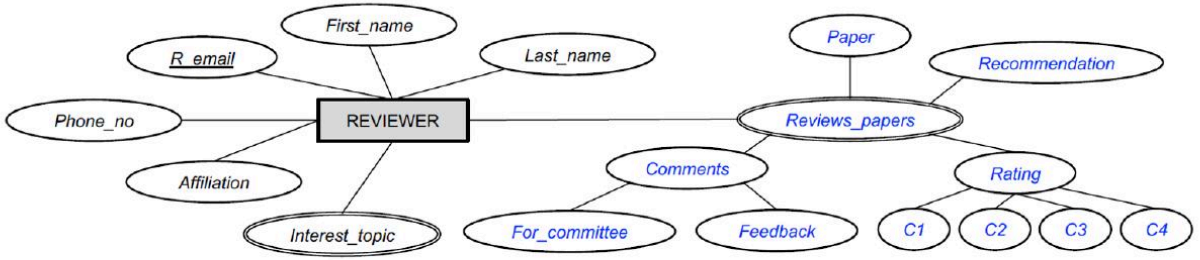
— 넘어가도 되는 쪽

Step 1: Describing the Entity Types (cont.)

❑ Entity Type 3: REVIEWER

■ Attributes (List entity types and their attributes based on the requirements):

R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic},
{Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee),
Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}



32

정답 Step 1: REVIEWER — 속성

```
R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic},
{Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee),
  Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}
```

- Interest_topic = 다중값 (이중 타원)
- Comments, Rating = 복합 속성

✍ 메모

★ 시험 Rating(C1~C4), Comments(Feedback, For_committee) = 복합 속성 2개

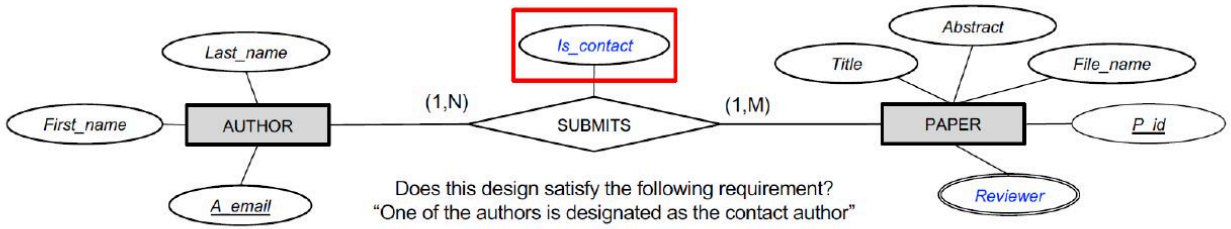
Step 2: Describing the Relationship Types

❑ Relationship Type 1: SUBMITS

■ Relating **AUTHOR** and **PAPER** entity types

- ❑ An author is required to submit a paper and a paper has to be submitted by an author to be recorded
  → both AUTHOR and PAPER types have a total participation (the minimum cardinality = 1)
- ❑ Author can submit many papers and a paper can be submitted by multiple authors (N:M)

■ List and describe relationship types and their cardinality ratios:



33

정답 Step 2: SUBMITS

- 저자는 논문을 제출해야, 논문은 저자가 제출해야 기록 → **AUTHOR, PAPER 둘 다 전체 참여** (최소 = 1)
- 저자는 여러 논문, 논문은 여러 저자 → **N:M**
- 그림: AUTHOR (1,N) — SUBMITS — (1,M) PAPER, 관계 속성 Is_contact
  - Q. 이 설계가 '저자 중 한 명이 교신 저자'라는 요구를 만족하나?

✍ 메모

⚠ 주의 Is_contact를 SUBMITS 속성으로 두면 '논문마다 정확히 한 명'을 강제 못 함 → 별도 관계 IS_CONTACT (다음 쪽)

Step 2: Describing the Relationship Types

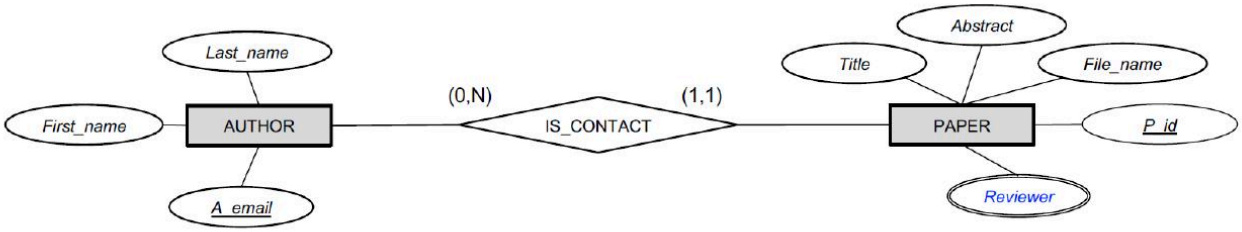
❑ Relationship Type 2: IS_CONTACT

■ Relating **AUTHOR** and **PAPER** entity types

❑ For each paper, one author should play the role of "the contact author"
→ PAPER has a total participation (the minimum cardinality = 1)

❑ An author can be the contact for many papers but does not have to be the contact for any papers
→ AUTHOR has a partial participation (the minimum cardinality = 0)

■ List and describe relationship types and their cardinality ratios:



34

정답 Step 2: IS_CONTACT

- 논문마다 저자 한 명이 교신 저자 → **PAPER 전체 참여** (최소 = 1)
- 저자는 여러 논문의 교신 저자일 수 있지만 아니어도 됨 → **AUTHOR 부분 참여** (최소 = 0)
- 그림: AUTHOR **(0,N)** — IS_CONTACT — **(1,1)** PAPER

✍ 메모

★ 시험 PAPER (1,1) = 교신 저자는 딱 한 명

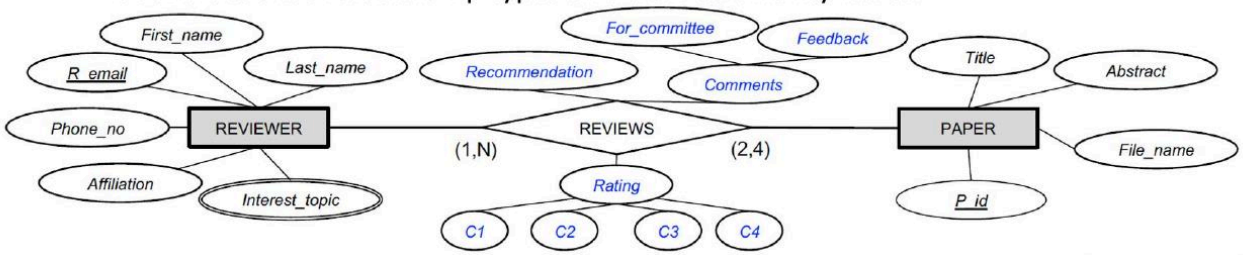
Step 2: Describing the Relationship Types

❑ Relationship Type 3: REVIEWS

■ Relating **REVIEWER** and **PAPER** entity types

- ❑ It has three relationship attributes including two composite attributes (*Comments* and *Rating*)
- ❑ A paper is reviewed by 2 to 4 reviewers
  → its (min, max) constraint is (2,4) (i.e., total participation)
- ❑ Every reviewer has to review at least one paper to be recorded and can review many papers
  → its (min, max) constraint is (1,N) (i.e., total participation)

■ List and describe relationship types and their cardinality ratios:

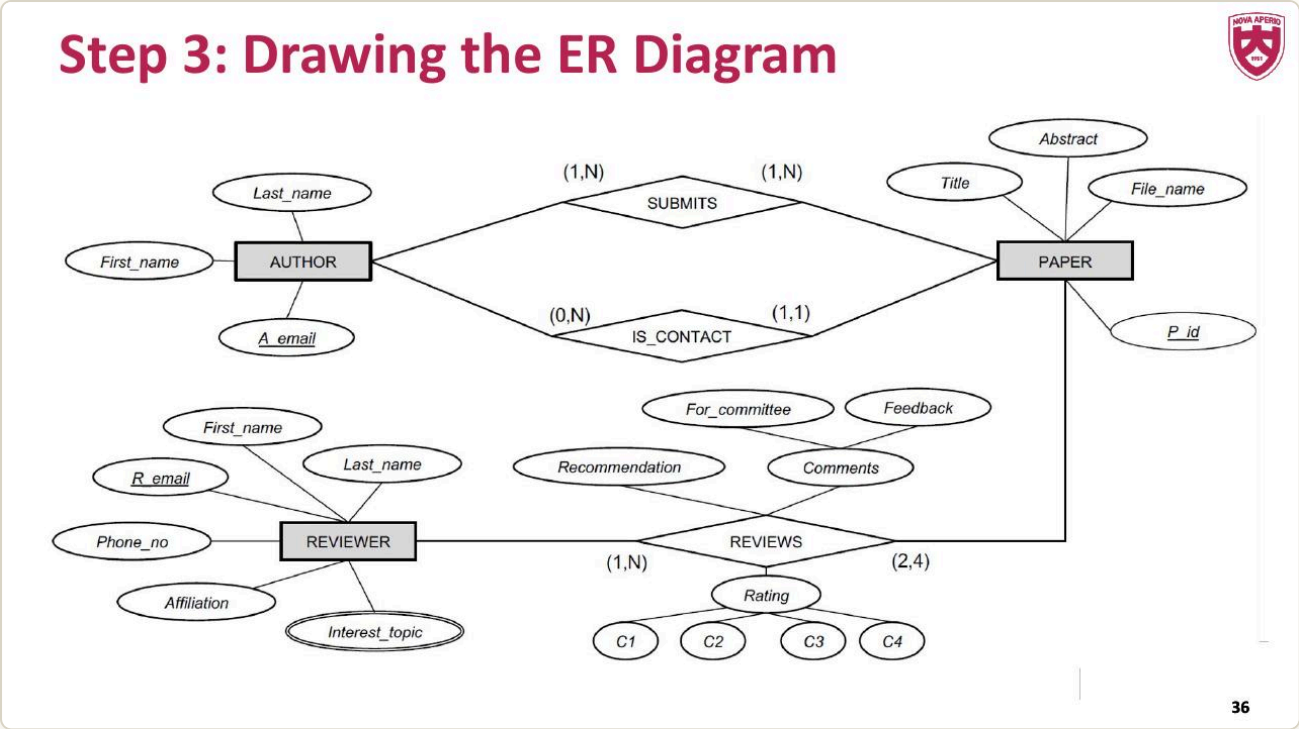


정답 Step 2: REVIEWS

- 관계 속성 3개 — 복합 2개(Comments, Rating) + Recommendation
- 논문은 심사위원 2~4명 → PAPER **(2,4)** (전체 참여)
- 심사위원은 최소 한 편, 여러 편 가능 → REVIEWER **(1,N)** (전체 참여)
- 그림: REVIEWER (1,N) — REVIEWS — (2,4) PAPER, Rating(C1~C4), Comments(For_committee, Feedback)

✍ 메모

★ 시험 (min,max)는 그 엔티티 옆: '논문은 2~4명' → PAPER 쪽 (2,4)



Step 3: 완성 ER 다이어그램 (CONFERENCE)

- AUTHOR(A_email, First_name, Last_name)
- PAPER(P_id, Title, Abstract, File_name)
- REVIEWER(R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic})
- AUTHOR (1,N) — **SUBMITS** — (1,N) PAPER
- AUTHOR (0,N) — **IS_CONTACT** — (1,1) PAPER
- REVIEWER (1,N) — **REVIEWS** — (2,4) PAPER, 관계 속성 Recommendation, Comments(...), Rating(C1~C4)

✍ 메모

★ 시험 정답 요약: SUBMITS M:N 양쪽 total · IS_CONTACT AUTHOR (0,N)·PAPER (1,1) · REVIEWS REVIEWER (1,N)·PAPER (2,4)